

Arya Patel

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EDUCATION

University of Waterloo

Bachelor of Computer Science and Finance

Waterloo, ON

2023 - 2027

TECHNICAL SKILLS

Languages: Python, Java, C++, C, SQL, JavaScript, TypeScript, HTML, CSS, Bash

Frameworks: Node.js, Flask, GraphQL, Next.js, React, TensorFlow, PyTorch, .NET Framework, Jest

Tools: MySQL, PostgreSQL, Docker, Kubernetes, AWS (RDS, S3, Lambda, EC2), GCP, SQL Server, GitLab CI/CD

EXPERIENCE

Municipal Property Assessment Corporation

Software Engineering Intern

May 2024 - Aug. 2024

- Optimized their **Python** based Geo-spatial **Convolutional Neural Network's** classification accuracy by **16%**
- Developed **4+** internal **RAG LLM** models, and reduced cloud environment response times by **0.4** seconds
- Built **AWS SQS** queue with **Django** endpoints, to automate malicious content checks within chatbot responses

Hack The North

Frontend Software Engineer

Jan. 2024 - Present

- Upgraded sponsorship dashboard to reduce data rendering time by **34%** in HTN's **distributed cloud** setup
- Implemented a **responsive** design on the dashboard table, allowing seamless usability across **10+** device types
- Built an automated reminder system, securing **\$500,000+** in sponsorship and reducing manual workload by **20%**

Guardian

Software Engineering Intern

Sept. 2023 - Feb. 2024

- Engineered a **Python backend** health appointment scheduling feature, driving app adoption to **600+ MAUs**
- Optimized scheduling UX with efficient **RESTful API** endpoints and **PostgreSQL** by reducing response times
- Improved app security protocols by automating regular vulnerability assessments using **AES-256 encryption**

PROJECTS

Echo | Hume AI, React, Node.js, JavaScript, TypeScript

- Developed a **real-time sentiment analysis** application for **reminiscence therapy** within dementia patients
- Outperformed industry benchmarks with a **25%** improvement in mood prediction accuracy within patient trials
- Implemented the **Hume AI API** to determine voice intonation and movement, for nuanced emotional analysis

Robo-Advisor | Python, Pandas, NumPy, Matplotlib, Git

- A **quantitative analysis script** predicting the least volatile portfolio from a set of arbitrary user-provided stocks
- Implemented **multi-threading**, queues and recursion to reduce request runtime from **RESTful APIs** by **53%**
- Consistently generated portfolios with **over 40% less volatility** when compared to the S&P500 and NASDAQ

Promptopia | JavaScript, TypeScript, Next.js, HTML/CSS, Git

- Developed a full-stack **CRUD** application using Next.js, utilizing **React Hooks** and **GraphQL** to streamline UX
- Achieved a **20%** reduction in page load times through **code refactoring**, allowing for exceptional app scalability
- Integrated an **OOP-based** framework utilizing **Java** and **SQL** for secure user data storage and retrieval processes

Movieland | JavaScript, React, OMDB API, HTML/CSS

- Reverse-engineered** the Netflix homepage to create a **React-based** clone with searchable movies and TV shows
- Featured **IMDb ratings**, enabling users to see public ratings for enhanced decision-making and user engagement